Name: _____

WORKSHEET

Complementary angle properties

Remember that complementary angles add to 90° or $\frac{\pi}{2}$. This table shows the relationship between equivalent sine and cosine angles according to complementary angle properties.

1st quadrant	2nd quadrant	3rd quadrant	4th quadrant
$\sin\left(\frac{\pi}{2}-\theta\right)=\cos(\theta)$ $\sin(90°-\theta)=\cos(\theta)$	$\sin\left(\frac{\pi}{2}+\theta\right)=\cos(\theta)$ $\sin(90°+\theta)=\cos(\theta)$	$\sin\left(\frac{3\pi}{2}-\theta\right)=-\cos(\theta)$ $\sin(270°-\theta)=-\cos(\theta)$	$\sin\left(\frac{3\pi}{2}+\theta\right)=-\cos(\theta)$ $\sin(270°+\theta)=-\cos(\theta)$
$\cos\left(\frac{\pi}{2}-\theta\right)=\sin(\theta)$ $\cos(90°-\theta)=\sin(\theta)$	$\cos\left(\frac{\pi}{2}+\theta\right)=-\sin(\theta)$ $\cos(90°+\theta)=-\sin(\theta)$	$\cos\left(\frac{3\pi}{2}-\theta\right)=-\sin(\theta)$ $\cos(270°-\theta)=-\sin(\theta)$	$\cos\left(\frac{3\pi}{2}+\theta\right)=\sin(\theta)$ $\cos(270°+\theta)=\sin(\theta)$
$\tan\left(\frac{\pi}{2}-\theta\right)=\cot(\theta)$ $\tan(90°-\theta)=\cot(\theta)$	$\tan\left(\frac{\pi}{2}+\theta\right)=-\cot(\theta)$ $\tan(90°+\theta)=-\cot(\theta)$	$\tan\left(\frac{3\pi}{2}-\theta\right)=\cot(\theta)$ $\tan(270°-\theta)=\cot(\theta)$	$\tan\left(\frac{3\pi}{2}+\theta\right)=-\cot(\theta)$ $\tan(270°+\theta)=-\cot(\theta)$

1 Use your understanding of complementary angles to write an equivalent angle for the following.

a $\cos(30°)$ _____

b $\tan(55°)$ _____

c $\sin(67°)$ _____

d $\sin\left(\frac{\pi}{4}\right)$ _____

e $\cos\left(\frac{\pi}{3}\right)$ _____

f $\tan\left(\frac{\pi}{6}\right)$ _____

2 Find the value of:

a $\sin(21°)$ if $\cos(69°)=0.3584$ _____

b $\cot(84°)$ if $\tan(6°)=0.1051$ _____

c $\cos(35°)$ if $\sin(55°)=0.8192$ _____

d $\tan(42°)$ if $\cot(48°)=0.9004$ _____

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3 Use your understanding of complementary angle properties to simplify the following.

a $\cos\left(\frac{\pi}{2} - \frac{\pi}{3}\right)$

b $\tan(90° - 60°)$

c $\tan\left(\frac{\pi}{2} - \frac{\pi}{6}\right)$

d $\cos\left(\frac{\pi}{2} + \frac{\pi}{4}\right)$

e $\sin(270° - 45°)$

f $\cos\left(\frac{3\pi}{2} + \frac{\pi}{6}\right)$

g $\sin\left(\frac{\pi}{2} + \frac{\pi}{3}\right)$

h $\cos(270° - 30°)$

i $\tan\left(\frac{\pi}{2} + \frac{\pi}{4}\right)$

j $\sin\left(\frac{3\pi}{2} + \frac{\pi}{6}\right)$

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4 If $\sin(x) = 0.25$, $\cos(\theta) = 0.58$ and $\tan(\beta) = 0.75$, find the value of the following:

a $\cos\left(\frac{\pi}{2} + x\right)$

b $\tan\left(\frac{3\pi}{2} - \beta\right)$

c $\cos\left(\frac{3\pi}{2} + x\right)$

d $\sin\left(\frac{\pi}{2} - \theta\right)$

e $\sin\left(\frac{3\pi}{2} - \theta\right)$

f $\tan\left(\frac{\pi}{2} - \beta\right)$

5 Rewrite each of these of these as complementary sums e.g. $\left(\frac{3\pi}{4} = \frac{\pi}{2} + \frac{\pi}{4}\right)$ and hence simplify using complementary angles properties and evaluate.

a $\sin\left(\frac{3\pi}{4}\right)$

b $\tan\left(\frac{4\pi}{3}\right)$

c $\sin\left(\frac{11\pi}{6}\right)$

d $\cos\left(\frac{2\pi}{3}\right)$

e $\cos\left(\frac{5\pi}{4}\right)$

f $\tan\left(\frac{5\pi}{6}\right)$

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Answers

1 **a** $\sin(60°)$

b $\cot(35°)$

c $\cos(23°)$

d $\cos\left(\frac{\pi}{4}\right)$

e $\sin\left(\frac{\pi}{6}\right)$

f $\cot\left(\frac{\pi}{3}\right)$

2 **a** 0.3584

b 0.1051

c 0.8192

d 0.9004

3 **a** $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

b $\cot(60°) = \frac{1}{\sqrt{3}}$

c $\cot\left(\frac{\pi}{6}\right) = \sqrt{3}$

d $-\sin\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

e $-\cos(45°) = -\frac{1}{\sqrt{2}}$

f $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$

g $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$

h $-\sin(30°) = -\frac{1}{2}$

i $-\cot\left(\frac{\pi}{4}\right) = -1$

j $-\cos\left(\frac{\pi}{6}\right) = -\frac{\sqrt{3}}{2}$

4 **a** -0.25

b $\frac{4}{3}$

c 0.25

d 0.58

e -0.58

f $\frac{4}{3}$

5 **a** $\sin\left(\frac{\pi}{2} + \frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

b $\tan\left(\frac{3\pi}{2} - \frac{\pi}{6}\right) = \cot\left(\frac{\pi}{6}\right) = \sqrt{3}$

c $\sin\left(\frac{3\pi}{2} + \frac{\pi}{3}\right) = -\cos\left(\frac{\pi}{3}\right) = \left(-\frac{1}{2}\right)$

d $\cos\left(\frac{\pi}{2} + \frac{\pi}{6}\right) = -\sin\left(\frac{\pi}{6}\right) = \left(-\frac{1}{2}\right)$

e $\cos\left(\frac{3\pi}{2} - \frac{\pi}{4}\right) = -\sin\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}}$

f $\tan\left(\frac{\pi}{2} + \frac{\pi}{3}\right) = -\cot\left(\frac{\pi}{3}\right) = -\frac{1}{\sqrt{3}}$

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